

Task 3: Memory Placement Algorithms

Task 3.1: Scenario 1

Initial State

At the given time, the following gaps exist in memory allocation (in memory order):

- $M1 = 9k$
- $M2 = 13k$
- $M3 = 52k$
- $M4 = 18k$
- $M5 = 15k$

The first process $P1 = 36k$ is allocated to the $M3$ gap ($52k$). After this allocation, the remaining free space in $M3$ is $52k - 36k = 16k$.

Updated free gaps after $P1$ allocation:

$M1 = 9k, \quad M2 = 13k, \quad M3 = 16k, \quad M4 = 18k, \quad M5 = 15k$

We now allocate the following four successive process requests: $P2 = 12k, P3 = 10k, P4 = 3k$, and $P5 = 8k$ using each of the four placement algorithms.

1. First Fit Algorithm

Algorithm description: First Fit places each process in the first available gap (in memory order) that is large enough to accommodate it.

Step-by-step allocation:

Allocate $P2 = 12k$:

- Check $M1 = 9k$: too small
- Check $M2 = 13k$: sufficient (first fit)
- Allocation: $P2 \rightarrow M2$
- Remaining in $M2$: $13k - 12k = 1k$
- Updated gaps:
 $M1 = 9k, M2 = 1k, M3 = 16k, M4 = 18k, M5 = 15k$

Allocate $P3 = 10k$:

- Check $M1 = 9k$: too small
- Check $M2 = 1k$: too small
- Check $M3 = 16k$: sufficient (first fit)
- Allocation: $P3 \rightarrow M3$
- Remaining in $M3$: $16k - 10k = 6k$
- Updated gaps:
 $M1 = 9k, M2 = 1k, M3 = 6k, M4 = 18k, M5 = 15k$

Allocate $P4 = 3k$:

- Check $M1 = 9k$: sufficient (first fit)
- Allocation: $P4 \rightarrow M1$
- Remaining in $M1$: $9k - 3k = 6k$
- Updated gaps:
 $M1 = 6k, M2 = 1k, M3 = 6k, M4 = 18k, M5 = 15k$

Allocate $P5 = 8k$:

- Check $M1 = 6k$: too small
- Check $M2 = 1k$: too small
- Check $M3 = 6k$: too small
- Check $M4 = 18k$: sufficient (first fit)
- Allocation: $P5 \rightarrow M4$
- Remaining in $M4$: $18k - 8k = 10k$
- Final gaps:
 $M1 = 6k, M2 = 1k, M3 = 6k, M4 = 10k, M5 = 15k$

First Fit Result: $P1-M3, P2-M2, P3-M3, P4-M1, P5-M4$

2. Best Fit Algorithm

Algorithm description: Best Fit places each process in the smallest available gap that is still large enough to accommodate it.

Step-by-step allocation:

Allocate $P2 = 12k$:

- Suitable gaps: $M2 = 13k, M3 = 16k, M4 = 18k, M5 = 15k$
- Smallest sufficient gap: $M2 = 13k$
- Allocation: $P2 \rightarrow M2$
- Remaining in $M2$: $13k - 12k = 1k$
- Updated gaps:
 $M1 = 9k, M2 = 1k, M3 = 16k, M4 = 18k, M5 = 15k$

Allocate $P3 = 10k$:

- Suitable gaps: $M3 = 16k, M4 = 18k, M5 = 15k$
- Smallest sufficient gap: $M5 = 15k$
- Allocation: $P3 \rightarrow M5$
- Remaining in $M5$: $15k - 10k = 5k$
- Updated gaps:
 $M1 = 9k, M2 = 1k, M3 = 16k, M4 = 18k, M5 = 5k$

Allocate $P4 = 3k$:

- Suitable gaps: $M1 = 9k, M3 = 16k, M4 = 18k, M5 = 5k$
- Smallest sufficient gap: $M5 = 5k$
- Allocation: $P4 \rightarrow M5$
- Remaining in $M5$: $5k - 3k = 2k$
- Updated gaps:
 $M1 = 9k, M2 = 1k, M3 = 16k, M4 = 18k, M5 = 2k$

Allocate $P5 = 8k$:

- Suitable gaps: $M1 = 9k, M3 = 16k, M4 = 18k$
- Smallest sufficient gap: $M1 = 9k$
- Allocation: $P5 \rightarrow M1$
- Remaining in $M1$: $9k - 8k = 1k$
- Final gaps:
 $M1 = 1k, M2 = 1k, M3 = 16k, M4 = 18k, M5 = 2k$

Best Fit Result: $P1-M3, P2-M2, P3-M5, P4-M5, P5-M1$

3. Next Fit Algorithm

Algorithm description: Next Fit continues searching from the position of the last successful allocation, maintaining a pointer to track the current search position. After placing $P1$ in $M3$, the search pointer begins at $M3$.

Step-by-step allocation:

Allocate $P2 = 12k$:

- Start search at $M3$ (last allocation position)
- Check $M3 = 16k$: sufficient
- Allocation: $P2 \rightarrow M3$
- Remaining in $M3$: $16k - 12k = 4k$
- Search pointer: remains at $M3$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 4k, M4 = 18k, M5 = 15k$

Allocate $P3 = 10k$:

- Start search at $M3$
- Check $M3 = 4k$: too small
- Move to $M4 = 18k$: sufficient
- Allocation: $P3 \rightarrow M4$
- Remaining in $M4$: $18k - 10k = 8k$
- Search pointer: moves to $M4$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 4k, M4 = 8k, M5 = 15k$

Allocate $P4 = 3k$:

- Start search at $M4$
- Check $M4 = 8k$: sufficient
- Allocation: $P4 \rightarrow M4$
- Remaining in $M4$: $8k - 3k = 5k$
- Search pointer: remains at $M4$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 4k, M4 = 5k, M5 = 15k$

Allocate $P5 = 8k$:

- Start search at $M4$
- Check $M4 = 5k$: too small
- Move to $M5 = 15k$: sufficient
- Allocation: $P5 \rightarrow M5$
- Remaining in $M5$: $15k - 8k = 7k$
- Search pointer: moves to $M5$
- Final gaps:
 $M1 = 9k, M2 = 13k, M3 = 4k, M4 = 5k, M5 = 7k$

Next Fit Result: $P1-M3, P2-M3, P3-M4, P4-M4, P5-M5$

4. Worst Fit Algorithm

Algorithm description: Worst Fit places each process in the largest available gap.

Step-by-step allocation:

Allocate $P2 = 12k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 18k, M5 = 15k$
- Largest gap: $M4 = 18k$
- Allocation: $P2 \rightarrow M4$
- Remaining in $M4$: $18k - 12k = 6k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 6k, M5 = 15k$

Allocate $P3 = 10k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 6k, M5 = 15k$
- Largest gap: $M3 = 16k$
- Allocation: $P3 \rightarrow M3$
- Remaining in $M3$: $16k - 10k = 6k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 6k, M4 = 6k, M5 = 15k$

Allocate $P4 = 3k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 6k, M4 = 6k, M5 = 15k$
- Largest gap: $M5 = 15k$
- Allocation: $P4 \rightarrow M5$
- Remaining in $M5$: $15k - 3k = 12k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 6k, M4 = 6k, M5 = 12k$

Allocate $P5 = 8k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 6k, M4 = 6k, M5 = 12k$
- Largest gap: $M2 = 13k$
- Allocation: $P5 \rightarrow M2$
- Remaining in $M2$: $13k - 8k = 5k$
- Final gaps:
 $M1 = 9k, M2 = 5k, M3 = 6k, M4 = 6k, M5 = 12k$

Worst Fit Result: $P1-M3, P2-M4, P3-M3, P4-M5, P5-M2$

Summary of Scenario 1 Results

Algorithm	Allocation Result
First Fit	$P1-M3, P2-M2, P3-M3, P4-M1, P5-M4$
Best Fit	$P1-M3, P2-M2, P3-M5, P4-M5, P5-M1$
Next Fit	$P1-M3, P2-M3, P3-M4, P4-M4, P5-M5$
Worst Fit	$P1-M3, P2-M4, P3-M3, P4-M5, P5-M2$

Task 3.2: Scenario 2

Initial State

The initial memory gaps are identical to Scenario 1:

$$M1 = 9k, \quad M2 = 13k, \quad M3 = 52k, \quad M4 = 18k, \quad M5 = 15k$$

Process $P1 = 36k$ is again allocated to $M3$, leaving $M3 = 16k$.

Updated free gaps after $P1$ allocation:

$$M1 = 9k, \quad M2 = 13k, \quad M3 = 16k, \quad M4 = 18k, \quad M5 = 15k$$

We now allocate the following four successive process requests:
 $P2 = 15k$, $P3 = 3k$, $P4 = 10k$, and $P5 = 8k$ using each of the four placement algorithms.

1. First Fit Algorithm

Step-by-step allocation:

Allocate $P2 = 15k$:

- Check $M1 = 9k$: too small
- Check $M2 = 13k$: too small
- Check $M3 = 16k$: sufficient (first fit)
- Allocation: $P2 \rightarrow M3$
- Remaining in $M3$: $16k - 15k = 1k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 1k, M4 = 18k, M5 = 15k$

Allocate $P3 = 3k$:

- Check $M1 = 9k$: sufficient (first fit)
- Allocation: $P3 \rightarrow M1$
- Remaining in $M1$: $9k - 3k = 6k$
- Updated gaps:
 $M1 = 6k, M2 = 13k, M3 = 1k, M4 = 18k, M5 = 15k$

Allocate $P4 = 10k$:

- Check $M1 = 6k$: too small
- Check $M2 = 13k$: sufficient (first fit)
- Allocation: $P4 \rightarrow M2$
- Remaining in $M2$: $13k - 10k = 3k$
- Updated gaps:
 $M1 = 6k, M2 = 3k, M3 = 1k, M4 = 18k, M5 = 15k$

Allocate $P5 = 8k$:

- Check $M1 = 6k$: too small
- Check $M2 = 3k$: too small
- Check $M3 = 1k$: too small

- Check $M4 = 18k$: sufficient (first fit)
- Allocation: $P5 \rightarrow M4$
- Remaining in $M4$: $18k - 8k = 10k$
- Final gaps:
 $M1 = 6k, M2 = 3k, M3 = 1k, M4 = 10k, M5 = 15k$

First Fit Result: $P1-M3, P2-M3, P3-M1, P4-M2, P5-M4$

2. Best Fit Algorithm

Step-by-step allocation:

Allocate $P2 = 15k$:

- Suitable gaps: $M3 = 16k, M4 = 18k, M5 = 15k$
- Smallest sufficient gap: $M5 = 15k$ (exact fit)
- Allocation: $P2 \rightarrow M5$
- Remaining in $M5$: $15k - 15k = 0k$ (gap eliminated)
- Updated gaps: $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 18k$

Allocate $P3 = 3k$:

- Suitable gaps: $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 18k$
- Smallest sufficient gap: $M1 = 9k$
- Allocation: $P3 \rightarrow M1$
- Remaining in $M1$: $9k - 3k = 6k$
- Updated gaps: $M1 = 6k, M2 = 13k, M3 = 16k, M4 = 18k$

Allocate $P4 = 10k$:

- Suitable gaps: $M2 = 13k, M3 = 16k, M4 = 18k$
- Smallest sufficient gap: $M2 = 13k$
- Allocation: $P4 \rightarrow M2$
- Remaining in $M2$: $13k - 10k = 3k$
- Updated gaps: $M1 = 6k, M2 = 3k, M3 = 16k, M4 = 18k$

Allocate $P5 = 8k$:

- Suitable gaps: $M3 = 16k, M4 = 18k$
- Smallest sufficient gap: $M3 = 16k$
- Allocation: $P5 \rightarrow M3$
- Remaining in $M3$: $16k - 8k = 8k$

- Final gaps: $M1 = 6k, M2 = 3k, M3 = 8k, M4 = 18k$

Best Fit Result: $P1-M3, P2-M5, P3-M1, P4-M2, P5-M3$

3. Next Fit Algorithm

Step-by-step allocation:

Allocate $P2 = 15k$:

- Start search at $M3$ (last allocation position)
- Check $M3 = 16k$: sufficient
- Allocation: $P2 \rightarrow M3$
- Remaining in $M3$: $16k - 15k = 1k$
- Search pointer: remains at $M3$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 1k, M4 = 18k, M5 = 15k$

Allocate $P3 = 3k$:

- Start search at $M3$
- Check $M3 = 1k$: too small
- Move to $M4 = 18k$: sufficient
- Allocation: $P3 \rightarrow M4$
- Remaining in $M4$: $18k - 3k = 15k$
- Search pointer: moves to $M4$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 1k, M4 = 15k, M5 = 15k$

Allocate $P4 = 10k$:

- Start search at $M4$
- Check $M4 = 15k$: sufficient
- Allocation: $P4 \rightarrow M5$
- Note: At this point, both $M4$ and $M5$ have 15k. Following memory order from search position, $M5$ comes after $M4$.
- Remaining in $M5$: $15k - 10k = 5k$
- Search pointer: moves to $M5$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 1k, M4 = 15k, M5 = 5k$

Allocate $P5 = 8k$:

- Start search at $M5$
- Check $M5 = 5k$: too small
- Wrap around to $M1 = 9k$: sufficient
- Allocation: $P5 \rightarrow M1$
- Remaining in $M1$: $9k - 8k = 1k$
- Final gaps:
 $M1 = 1k, M2 = 13k, M3 = 1k, M4 = 15k, M5 = 5k$

Next Fit Result: $P1-M3, P2-M3, P3-M4, P4-M5, P5-M1$

4. Worst Fit Algorithm

Step-by-step allocation:

Allocate $P2 = 15k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 18k, M5 = 15k$
- Largest gap: $M4 = 18k$
- Allocation: $P2 \rightarrow M4$
- Remaining in $M4$: $18k - 15k = 3k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 3k, M5 = 15k$

Allocate $P3 = 3k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 16k, M4 = 3k, M5 = 15k$
- Largest gap: $M3 = 16k$
- Allocation: $P3 \rightarrow M3$
- Remaining in $M3$: $16k - 3k = 13k$
- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 13k, M4 = 3k, M5 = 15k$

Allocate $P4 = 10k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 13k, M4 = 3k, M5 = 15k$
- Largest gap: $M5 = 15k$
- Allocation: $P4 \rightarrow M5$
- Remaining in $M5$: $15k - 10k = 5k$

- Updated gaps:
 $M1 = 9k, M2 = 13k, M3 = 13k, M4 = 3k, M5 = 5k$

Allocate $P5 = 8k$:

- Available gaps:
 $M1 = 9k, M2 = 13k, M3 = 13k, M4 = 3k, M5 = 5k$
- Largest gaps: $M2 = 13k$ and $M3 = 13k$ (tie)
- When multiple gaps are equal, select first in memory order: $M2$
- Allocation: $P5 \rightarrow M2$
- Remaining in $M2$: $13k - 8k = 5k$
- Final gaps:
 $M1 = 9k, M2 = 5k, M3 = 13k, M4 = 3k, M5 = 5k$

Worst Fit Result: $P1-M3, P2-M4, P3-M3, P4-M5, P5-M2$

Summary of Scenario 2 Results

Algorithm	Allocation Result
First Fit	$P1-M3, P2-M3, P3-M1, P4-M2, P5-M4$
Best Fit	$P1-M3, P2-M5, P3-M1, P4-M2, P5-M3$
Next Fit	$P1-M3, P2-M3, P3-M4, P4-M5, P5-M1$
Worst Fit	$P1-M3, P2-M4, P3-M3, P4-M5, P5-M2$

Conclusion

This task demonstrates the four fundamental memory placement algorithms used in operating systems. Each algorithm exhibits different characteristics in terms of allocation efficiency and memory fragmentation:

- **First Fit** offers fast allocation by selecting the first suitable gap, but may lead to fragmentation at the beginning of memory.
- **Best Fit** minimizes wasted space by selecting the smallest sufficient gap, though it can create many small unusable gaps.
- **Next Fit** distributes allocations more evenly across memory by continuing from the last allocation point, potentially reducing search time.

- **Worst Fit** leaves larger remaining gaps after allocation, which may accommodate future larger processes, but can lead to poor memory utilization overall.

The choice of algorithm depends on the specific memory access patterns and process size distributions in a given system.